

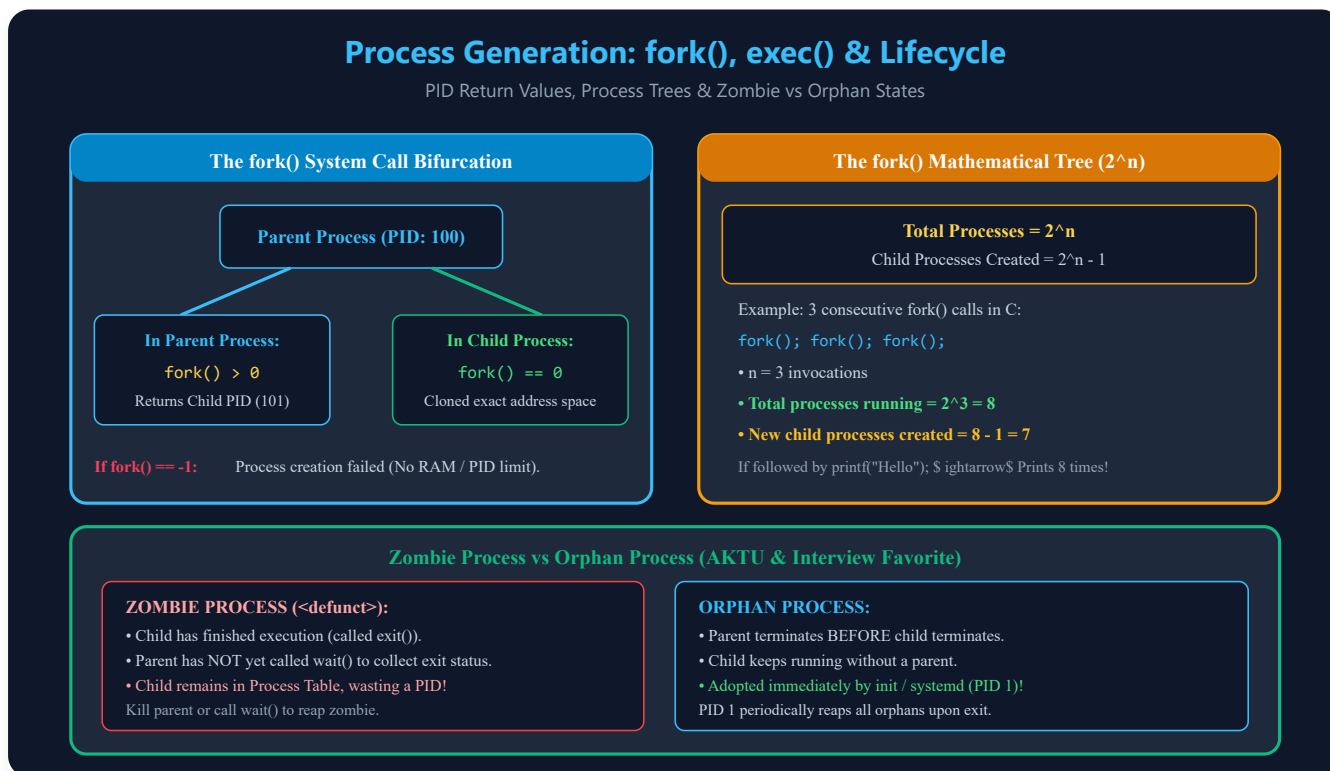
10. Process Generation, Forking & Lifecycle

Syllabus Coverage: Gateway Classes Slides 121–128 | Forking vs Spawning, Return Values, 2^n Tree Mathematics, Zombie vs Orphan Processes.

10.1 The fork() System Call

Clones the calling parent process into an identical child address space.

- `pid == 0` in Child process.
- `pid > 0` in Parent process (value is child PID).
- `pid == -1` on process creation failure.



10.2 Mathematical Process Tree Growth

Formula for n consecutive `fork()` statements:

Total Processes = 2^n
Total Child Processes Created = $2^n - 1$

10.3 Zombie vs Orphan Process Comparison

Feature	Zombie Process (<defunct>)	Orphan Process
Terminated First	Child terminated first; Parent still alive.	Parent terminated first; Child still running.
Process Table Entry	Entry remains in Process Table (wastes PID).	Active normal process entry.
Resolution / Adoption	Parent must call <code>wait()</code> to reap status.	Immediately adopted by <code>init</code> / <code>systemd</code> (PID 1).